

TechOps Inbound Specifications

For Longform Content

VERSION 2.8

March 2024

SUMMARY

Contained in this document you will find proper technical specifications and delivery instructions. If there are any questions regarding these guidelines, please contact NBCUniversal Technical Operations (TechOps). We look forward to bringing your project into TechOps!

USER GUIDE

The first portion of the document contains **General Tech Ops** specifications. The second portion of the document with **color coded tabs** contains guidelines specific to your delivery:

International SkyShowtime Network, Cable & Studio Direct-to-Broadcast Peacock

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General TechOps Guidelines

Onboarding Form

At the beginning of the onboarding process, TechOps will provide an onboarding form to the production or delivery partner. Within the onboarding form, the production/delivery partner will need to fill out the following:

- Video specs
- Audio specs
- Caption specs
- Presence of ad breaks
- Content delivery estimated timeline
- Delivery method
- Test file delivery
- Contact info

Onboarding Call

Once the onboarding form and all supplemental documents have been handed off and completed, TechOps will schedule an initial kick-off call to review the form's contents, the metadata document, and any additional questions regarding this spec document or the associated procedures. Please have your completed form, completed metadata document, and any additional questions prepared prior to the call.

General Audio Guidelines

Audio Requirements

DESCRIPTION	REQUIREMENT
Audio Spec	LPCM Little Endian Discrete Audio – 1 Channel per 1 Track
Audio Sampling Rate	48 kHz
Audio Bit Rate	24-bit
Audio Program Loudness for Broadcast	-24 LKFS
Audio Dynamic Limit for Broadcast	-2 dBFS True Peak
Audio Program Loudness for Masters	Original mix
Audio Dynamic Limit for Masters	As source
Audio Reference Level	-24 LKFS 1 kHz tone

I. AUDIO STANDARD

Calm Act

For CALM ACT Compliancy (United States Broadcast), normal dialogue, not whispered or shouted, must be maintained at -24LKFS as closely as possible. This will ensure that the loudness meets federal regulations, that there is not an issue with an episode's loudness & consistency in relation to ads and promos, and will make for a good viewer experience.

**A tight tolerance surrounding the -24LKFS average is especially critical at the start and end of content or segments to establish smooth program to commercial transitions.*

Dialog loudness: -24 LKFS	Dynamic limit: -2 dBFS True Peak	Loudness Range: -12 LKFS
----------------------------------	---	---------------------------------

EAS Tone

The FCC prohibits the transmission of the Emergency Alert System (EAS) codes or Attention Signal (853 Hz) simulation thereof except in the case of an actual emergency. The rule applies to all forms of content, including programs commercials and other paid programming, PSAs, promos and other interstitial material. The FCC has not defined what constitutes a prohibited "simulation" and has brought enforcement actions against sound effects other than actual EAS tones. Content containing EAS tones or simulations will not be accepted for broadcast.

I. AUDIO IDENTIFICATION

Audio should be configured with channel identification. This information should be available when using media info and can be added to a file a variety of ways. An example of audio channel identification for a 12-channel file is below:

Track	Name	File	Channels	Format	Language	Availability	Mute
A	Track 1	Test....mov	Left	PCM	English	AVAILABLE	☐
B	Track 2	Test....mov	Right	PCM	English	AVAILABLE	☐
C	Track 3	Test....mov	Center	PCM	English	AVAILABLE	☐
D	Track 4	Test....mov	LFE	PCM	English	AVAILABLE	☐
E	Track 5	Test....mov	Left Back	PCM	English	AVAILABLE	☐
F	Track 6	Test....mov	Right Back	PCM	English	AVAILABLE	☐
G	Track 7	Test....mov	Left Total	PCM	English	AVAILABLE	☐
H	Track 8	Test....mov	Right Total	PCM	English	AVAILABLE	☐
I	Track 9	Test....mov	Left Total	PCM	English	AVAILABLE	☐
J	Track 10	Test....mov	Right Total	PCM	English	AVAILABLE	☐
K	Track 11	Test....mov	Left Total	PCM	English	AVAILABLE	☐
L	Track 12	Test....mov	Right Total	PCM	English	AVAILABLE	☐

AUDIO DUBBING REQUIREMENTS

The following are required deliverables for dubbed languages.

- 5.1 Mixed Language Print Master **Interleaved WAV File**
- 2 Channel Left Total/Right Total (LT/RT) Mixed Language Print Master **Interleaved WAV File**

The following audio file and session specifications should be adhered to.

- Regarding Pro Tools sessions, all files must conform to the native framerate and video guide, matching in session length and synchronizing with picture

CREATING INTERLEAVED FILES

Using final Pro Tools Session Print Master material once saved for delivery, please prepare session as follows:

- Ensure that session settings and file start match the guide.
- Ensure that all channels are completely consolidated with no edits.
- Ensure that channels are configured in the proper channel order.
 - 5.1 = L, C, R, LS, RS, LFE
 - 2.0 = LT, RT
- If channels are in single mono tracks, create new single 5.1 and stereo tracks and drag audio files in order to the new 5.1 and stereo tracks.
- Remove/do not include tail sync pops on any .wav files.

From the prepared session, export interleaved files in accordance with the following:

- Select/highlight entire 5.1 and LT/RT track(s).
- In the Clips section, click on “Export Clips as Files.”
- In the Export Selected pop-up menu, ensure settings:
 - File Type is WAV
 - Format is Interleaved
 - Bit Depth is 24 Bit
 - Sample Rate is 48kHz
- Click Export and check that single files are being created for each channel group (5.1, LT/RT, etc.).
- Final audio layout of interleaved file must be:
 - 5.1 = L, R, C, LFE, LS, RS
 - 2.0 = LT, RT

*If Pro Tools session is set in “film mode” (L, C, R, LS, RS, LFE), when files are exported to an interleaved file, they will be in SMPTE layout (L, R, C, LFE, LS, RS).

General Component Requirements

The guidelines and table below outline the requirements for component delivery.

Please note that all component deliveries should be **delivered as sidecar** and **conformed to the Video Master file (i.e., all components should be in sync with the video)**.

TEXT/CAPTION COMPONENT GUIDELINES

- Captions **must have a period at the end** of each segment. If there is no period, captions will remain on-screen
- All closed captioning, subtitle, or forced narrative positioning should **not** cover actor faces, on-screen graphics, or on-screen text.
- Bold, italics, or other stylizations are **not** recommended for subtitle files (and if included may not appear as intended)
- **Positional data** should be included to ensure subtitle elements do not overlap any on-screen text
- Only **UTF-8 encoding** is accepted
- Remember to match framerates
 - **29.97** content must be **Drop Frame**
 - **23.98** and **25** must be **Non-Drop Frame**
- Captions **must remain onscreen** for a minimum of **10 frames**
- **Time code** should be expressed in frames in the **HH:MM:SS:FF** format (no milliseconds or absolute frame count).
- All subtitle deliverables intended to be used to support file fulfillment, shall conform to the IMSC 1.1 spec as defined by the W3C and found here:
 - <https://www.w3.org/TR/ttml-ismc1.1/>
- For **IMSC1.1** files, timing reference should be set to **"MEDIA"** to match the time code data track of the video asset to which the component is conformed.
 - Please see following section for detailed specs on the creation of IMSC 1.1 Subtitles for HD or UHD deliverables.
- **Line breaks** should be placed according to proper **grammatical breaks** and **natural phrasing**
- **Special characters** should all be **correctly coded** to the specifications of the caption format
- **Captions** should be **consistent with audio**. If profanity is **audible**, it should be included in the captions; if profanity or slang have been **censored**, this should be indicated in the captions.
- **Sound effects** should be captioned when necessary to the understanding of the media
- Captions for **music** should include references to the **title** and **artist** whenever possible
- **Lyrics** should be enclosed by **musical notes** at the beginning and end of each line
- Captions sent as **SCC files** can include up to **three lines of text** per timecode and can only have **32 characters MAX** per line.
 - Note: If **Italics** are being used then the **MAX characters per line is 31**. Words wither need to be moved to the next row (to have 3 rows on screen instead of 2) or moved down to the next text box.
- Text may be mixed case or upper case, but **should remain consistent** throughout each episode, with exceptions for emphasis or off-screen narration/dialogue

IMSC 1.1 Subtitle Specifications

File Delivery Specification for UHD / HDR IMSC 1.1

All subtitle deliverables intended to be used to support Servicing shall conform to the IMSC-1.1 spec as defined by the W3C and found here: <https://www.w3.org/TR/ttml-imsc1.1/>

- The Data Essence shall conform to the IMSC-1.1 “Text” Profile and include all spoken word and text events that appear on screen.
- IMSC-1.1 Character based language formatting is listed below
- IMSC-1.1 “PNG” Profile is **not acceptable** as a deliverable for any language.
- File format extension **must be delivered in .TTML**
- Please note: NBCUniversal will **reject** files delivered with the extension .XML

File Format Specifications (Data) UHD/HDR

- Position, Size, Luminance / Color Tag information etc. should be consistent for the content across all versions of the same Film or TV content.
 - *Example – Theatrical version, Textless Version, Directors Version, Edited Version etc. should be consistent in regard to the “look” of the IMSC-1 Titles*
- Consistency need not be maintained from movie to movie or TV show to TV show, only within the given title or series.
- Text “look” should be consistent with the theatrical and remain consistent version to version (i.e. Theatrical, Directors cut, German, French etc.)
- Frame Rate and Multiplier **must** match Time Code (time base) of the file
- Sizing should be adjusted for non-theatrical use. 80% adjustment from theatrical acceptable for most languages.
- Character based languages require 100% sizing in order to read font (i.e. Cantonese (Hong Kong), Japanese, Korean, Arabic, Trad Mandarin (Taiwan), Thai)
- Text alignment / positioning should be either left/right or center to match the original aspect ratio version of the content (matching the theatrical version)
- Luminance Gain and Border information **must be present** for IMSC -1.1 Data deliverables:
- Luminance Gain should be **120 nit** or a **factor of 1.5** should this file ever be used for HDR purposes.
- Example of “luminance gain” listed in XML <region> area.
- The Border **must** be black
- Grey or any other color will trigger a **rejection**
- NBCUniversal defines black for these titles as “Full” range
- Therefore the value in the XML **should be #00000**
- Representing 0/1023 or 0/255 code values
- The Header information should be the similar (depending on the tool used to create) as the XML format example below:
- Example for PQ (UHD/HDR):

```
<?xml version="1.0" encoding="utf-8"?>
<tt xmlns="http://www.w3.org/ns/ttml" xmlns:ttp="http://www.w3.org/ns/ttml#parameter"
xmlns:tts="http://www.w3.org/ns/ttml#styling" xmlns:ttm="http://www.w3.org/ns/ttml#metadata"
xmlns:ebutts="urn:ebu:tt:style" xmlns:ebuttm="urn:ebu:tt:metadata" xml:lang="sk"
ttp:contentProfiles="http://www.w3.org/ns/ttml/profile/imsc1.1/text" ttp:timeBase="media" ttp:frameRate="24"
ttp:frameRateMultiplier="1000 1001">
  <head>
    <metadata>
      <ttm:title>GTMV0000111111_04444_SUB_HL-PQ</ttm:title>
      <ttm:desc>IMSC 1.1 Text Profile document created by Sferastudios</ttm:desc>
      <ebuttm:conformsToStandard>urn:ebu:tt:distribution:2014-01</ebuttm:conformsToStandard>
      <ebuttm:conformsToStandard>urn:ebu:tt:distribution:2014-01</ebuttm:conformsToStandard>
      <ebuttm:conformsToStandard>http://www.w3.org/ns/ttml/profile/imsc1/text</ebuttm:conformsToStandard>
      <ebuttm:conformsToStandard>http://www.w3.org/ns/ttml/profile/imsc1.1/text</ebuttm:conformsToStandard>
    </metadata>
```

- The styling and layout information should appear as below:
- Example for PQ (UHD/HDR)
- Font color is 127 grey 127,127,127 #7F7F7F with full range black outline 000000 at 3% thick

```
<styling>
  <style xml:id="default" tts:color="#7f7f7f" tts:backgroundColor="transparent"
tts:fontFamily="proportionalSansSerif" tts:fontSize="80%" tts:fontWeight="normal" tts:fontStyle="normal"
tts:textOutline="#000000 3%"/>
</styling>
<layout>
  <region xml:id="Region_Top" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="center" tts:displayAlign="before" tts:luminanceGain="1.5"/>
  <region xml:id="Region_Bottom" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="center" tts:displayAlign="after" tts:luminanceGain="1.5"/>
  <region xml:id="Region_Vertical_Left" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="start" tts:displayAlign="after" tts:writingMode="tbrl"
tts:luminanceGain="1.5"/>
  <region xml:id="Region_Vertical_Right" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="start" tts:displayAlign="before" tts:writingMode="tbrl"
tts:luminanceGain="1.5"/>
</layout>
</head>
```

- The position, luminance / color tag, and time base information etc. should be in the format below:
- Example for PQ (UHD/HDR):
- **Note:** the default style and subsequent italic conversions on a particular line if needed highlighted for reference

```
<body>
  <div>
    <p begin="01:01:16:11" end="01:01:19:19" style="default" region="Region_Bottom">PRED DVOMA DŇAMI</p>
    <p begin="01:01:58:10" end="01:01:59:11" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Brendan?</span></p>
    <p begin="01:02:03:16" end="01:02:04:16" style="default" region="Region_Bottom">Emily?</p>
    <p begin="01:02:06:04" end="01:02:07:04" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Áno.</span></p>
```

```

<p begin="01:02:09:06" end="01:02:10:16" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Ako sa máš?</span></p>
<p begin="01:02:13:12" end="01:02:15:01" style="default" region="Region_Bottom">Stále rovnako.</p>
<p begin="01:02:16:14" end="01:02:18:04" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Áno?</span></p>
<p begin="01:02:20:21" end="01:02:22:01" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">To je fajn.</span></p>
<p begin="01:02:30:17" end="01:02:32:09" style="default" region="Region_Bottom">Čo je, Emily?</p>
<p begin="01:02:32:11" end="01:02:36:07" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Rada ťa počujem, Brendan.</span></p>
<p begin="01:02:38:17" end="01:02:40:12" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Dlho sme sa nevideli.</span></p>
<p begin="01:02:41:18" end="01:02:42:21" style="default" region="Region_Bottom">Dva mesiace.</p>
<p begin="01:02:43:06" end="01:02:47:22" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Áno, zabudla som číslo</span><br/><span tts:fontStyle="italic">tvojej skrinky. Povedal mi ho
Chytrák.</span></p>
<p begin="01:02:48:17" end="01:02:50:24" style="default" region="Region_Bottom">Čo sa niekde stretnúť?</p>
<p begin="01:02:52:09" end="01:02:53:12" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Nemôžem.</span></p>
<p begin="01:02:54:07" end="01:02:58:03" style="default" region="Region_Bottom">-Prečo nie?<br/><span
tts:fontStyle="italic">-Ja som to pobabrala.</span></p>
<p begin="01:02:59:04" end="01:03:01:04" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Poriadne.</span></p>
<div>
<body>

```

FILE FORMAT SPECIFICATIONS (DATA) for Character Based Languages that may contain horizontal and vertical alignment concurrently (i.e., Japan)

The Header information should be the similar (depending on the tool used to create) as the XML format example below:

- Example for PQ (UHD/HDR):

```

<?xml version="1.0" encoding="utf-8"?>
<tt xmlns="http://www.w3.org/ns/ttml" xmlns:ttp="http://www.w3.org/ns/ttml#parameter"
xmlns:tts="http://www.w3.org/ns/ttml#styling" xmlns:ttm="http://www.w3.org/ns/ttml#metadata"
xmlns:ebutts="urn:ebu:tt:style" xmlns:ebuttm="urn:ebu:tt:metadata" xml:lang="ja"
ttp:contentProfiles="http://www.w3.org/ns/ttml/profile/imsc1.1/text" ttp:timeBase="media" ttp:frameRate="24"
ttp:frameRateMultiplier="1000 1001">
<head>
<metadata>
<ttm:title>GTMV0000100000_04001_SUB_JA-JA</ttm:title>
<ttm:desc>IMSC 1.1 Text Profile document created by Sferastudios</ttm:desc>
<ebuttm:conformsToStandard>urn:ebu:tt:distribution:2014-01</ebuttm:conformsToStandard>
<ebuttm:conformsToStandard>urn:ebu:tt:distribution:2014-01</ebuttm:conformsToStandard>
<ebuttm:conformsToStandard>http://www.w3.org/ns/ttml/profile/imsc1.1/text</ebuttm:conformsToStandard>
<ebuttm:conformsToStandard>http://www.w3.org/ns/ttml/profile/imsc1.1/text</ebuttm:conformsToStandard>
</metadata>

```

- The styling and layout information should appear as below:
- Example for PQ (UHD/HDR)

- Font color is 127 grey 127,127,127 #7F7F7F with full range black outline 000000 at 3% thick. Font size differences highlighted as noted in section 2.0.4a above.

```
<styling>
  <style xml:id="default" tts:color="#7f7f7f" tts:backgroundColor="transparent"
tts:fontFamily="proportionalSansSerif" tts:fontSize="100%" tts:fontWeight="normal" tts:fontStyle="normal"
tts:textOutline="#000000 3%"/>
</styling>
<layout>
  <region xml:id="Region_Top" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="center" tts:displayAlign="before" tts:luminanceGain="1.5"/>
  <region xml:id="Region_Bottom" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="center" tts:displayAlign="after" tts:luminanceGain="1.5"/>
  <region xml:id="Region_Vertical_Left" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="start" tts:displayAlign="after" tts:writingMode="tbrl"
tts:luminanceGain="1.5"/>
  <region xml:id="Region_Vertical_Right" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="start" tts:displayAlign="before" tts:writingMode="tbrl"
tts:luminanceGain="1.5"/>
</layout>
```

- The position, luminance / color tag, and time base information etc. should be in the format below:
- Example for PQ (UHD/HDR):
- Note:** the default style and subsequent vertical alignment with shear adjustment example on a particular line if needed are highlighted for reference

```
</head>
<body>
  <div>
    <p begin="01:26:01:16" end="01:26:02:19" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">カ  
メラが先だ</p>
    <p begin="01:27:28:10" end="01:27:32:09" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">他  
人は私を<br/>トレジャーハンターと呼ぶ</p>
    <p begin="01:27:32:11" end="01:27:34:02" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">で  
も誤解よ</p>
    <p begin="01:27:34:05" end="01:27:36:14" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">金  
銭が目的じゃない</p>
    <p begin="01:27:37:15" end="01:27:40:11" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">私  
は錬金術の 歴史の研究者よ</p>
    <p begin="01:27:42:15" end="01:27:43:19" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">あ  
の石には&#11834;</p>
    <p begin="01:27:44:21" end="01:27:46:19" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">歴  
史的価値がある</p>
    <p begin="01:27:48:07" end="01:27:49:23" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">真  
実の探求が目的？</p>
    <p begin="01:27:50:07" end="01:27:51:07" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">そ  
うよ</p>
    <p begin="01:27:54:17" end="01:27:56:06" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">真  
実が知りたい</p>
    <p begin="01:32:13:01" end="01:32:15:19" style="default" region="Region_Vertical_Right" tts:shear="16.67%">出るに  
は下りるしかない</p>
  </div>
</body>
</tt>
```

File Delivery Specification for HD / SDR IMSC 1.1

All closed captions and subtitle deliverables intended to be used to support NBCUniversal's GMO Servicing library shall conform to the IMSC-1.1 spec as defined by the W3C and found here: <https://www.w3.org/TR/ttml-imsc1.1/>

- The Data Essence shall conform to the IMSC-1.1 "Text" Profile and include all spoken word and text events that appear on screen.
- IMSC-1.1 Character based language formatting is listed in section 3.0
- IMSC-1.1 "PNG" Profile is **not acceptable** as a deliverable for any language.
- File format extension **must be delivered in .TTML**
 - Please note: NBCUniversal will **reject** files delivered with the extension .XML

FILE FORMAT SPECIFICATIONS (DATA)

- Position, Size, Luminance / Color Tag information etc. should be consistent for the content across all versions of the same Film or TV content.
 - *Example – Theatrical version, Textless Version, Directors Version, Edited Version etc. should be consistent in regard to the "look" of the IMSC-1 Titles*
- Consistency need not be maintained from movie to movie or TV show to TV show, only within the given title or series.
- Text "look" should be consistent with the theatrical and remain consistent version to version (i.e. Theatrical, Directors cut, German, French etc.)
- Frame Rate and Multiplier **must** match Time Code (time base) of the file
- Sizing should be adjusted for non-theatrical use. 80% adjustment from theatrical acceptable for most languages.
- Character based languages require 100% sizing in order to read font (i.e. Cantonese (Hong Kong), Japanese, Korean, Arabic, Trad Mandarin (Taiwan), Thai)
- Text alignment / positioning should be either left/right or center to match the original aspect ratio version of the content (matching the theatrical version)
- Luminance Gain and Border information **must be present** for IMSC -1.1 Data deliverables:
- Luminance Gain should be **120 nit** or a **factor of 1.5** should this file ever be used for HDR purposes.
- Example of "luminance gain" listed in XML <region> area.
- The Border **must** be black
- Grey or any other color will trigger a **rejection**
- NBCUniversal defines black for these titles as "legal/limited/narrow" range, not "Full" range
- Therefore the value in the XML **should be #101010** vs. #000000
- Representing 64/940 or 16/235 code values
- The Header information should be the similar (depending on the tool used to create) as the XML format example below:

- Example for 709 (HD/SDR):

```
<?xml version="1.0" encoding="utf-8"?>
<tt xmlns="http://www.w3.org/ns/ttml" xmlns:ttp="http://www.w3.org/ns/ttml#parameter"
xmlns:tts="http://www.w3.org/ns/ttml#styling" xmlns:ttm="http://www.w3.org/ns/ttml#metadata"
xmlns:ebutts="urn:ebu:tt:style" xmlns:ebuttm="urn:ebu:tt:metadata" xml:lang="sk"
ttp:contentProfiles="http://www.w3.org/ns/ttml/profile/imsc1.1/text" ttp:timeBase="media"
ttp:frameRate="24" ttp:frameRateMultiplier="1000 1001">
<head>
<metadata>
<ttm:title>GTMV0000102538_04024_SUB_SK-SK</ttm:title>
<ttm:desc>IMSC 1.1 Text Profile document created by Sferastudios</ttm:desc>
<ebuttm:conformsToStandard>urn:ebu:tt:distribution:2014-01</ebuttm:conformsToStandard>
<ebuttm:conformsToStandard>urn:ebu:tt:distribution:2014-01</ebuttm:conformsToStandard>
<ebuttm:conformsToStandard>http://www.w3.org/ns/ttml/profile/imsc1/text</ebuttm:conformsToStandard>
<ebuttm:conformsToStandard>http://www.w3.org/ns/ttml/profile/imsc1.1/text</ebuttm:conformsToStandard>
</metadata>
```

- The styling and layout information should appear as below:
- Example for 709 (HD/SDR):
- Font color is 92 grey 235,235,235 with legal range black outline 101010 at 3% thick

```
<styling>
<style xml:id="default" tts:color="#ebeb33" tts:backgroundColor="transparent"
tts:fontFamily="proportionalSansSerif" tts:fontSize="80%" tts:fontWeight="normal" tts:fontStyle="normal"
tts:textOutline="#101010 3%"/>
</styling>
<layout>
<region xml:id="Region_Top" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="center" tts:displayAlign="before" tts:luminanceGain="1.5"/>
<region xml:id="Region_Bottom" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="center" tts:displayAlign="after" tts:luminanceGain="1.5"/>
<region xml:id="Region_Vertical_Left" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="start" tts:displayAlign="after" tts:writingMode="tbrl"
tts:luminanceGain="1.5"/>
<region xml:id="Region_Vertical_Right" tts:backgroundColor="transparent" tts:showBackground="whenActive"
tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="start" tts:displayAlign="before" tts:writingMode="tbrl"
tts:luminanceGain="1.5"/>
</layout>
</head>
```

- The position, luminance / color tag, and time base information etc. should be in the format below:
- Example for 709 (HD/SDR):
- **Note:** the default style and subsequent italic conversions on a particular line if needed highlighted for reference

```
<body>
<div>
<p begin="01:01:16:11" end="01:01:19:19" style="default" region="Region_Bottom">PRED DVOMA DŇAMI</p>
<p begin="01:01:58:10" end="01:01:59:11" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Brendan?</span></p>
<p begin="01:02:03:16" end="01:02:04:16" style="default" region="Region_Bottom">Emily?</p>
<p begin="01:02:06:04" end="01:02:07:04" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Áno.</span></p>
<p begin="01:02:09:06" end="01:02:10:16" style="default" region="Region_Bottom"><span tts:fontStyle="italic">Ako
sa máš?</span></p>
```

```

<p begin="01:02:13:12" end="01:02:15:01" style="default" region="Region_Bottom">Stále rovnako.</p>
<p begin="01:02:16:14" end="01:02:18:04" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Áno?</span></p>
<p begin="01:02:20:21" end="01:02:22:01" style="default" region="Region_Bottom"><span tts:fontStyle="italic">To je
fajn.</span></p>
<p begin="01:02:30:17" end="01:02:32:09" style="default" region="Region_Bottom">Čo je, Emily?</p>
<p begin="01:02:32:11" end="01:02:36:07" style="default" region="Region_Bottom"><span tts:fontStyle="italic">Rada
ťa počujem, Brendan.</span></p>
<p begin="01:02:38:17" end="01:02:40:12" style="default" region="Region_Bottom"><span tts:fontStyle="italic">Dlho
sme sa nevideli.</span></p>
<p begin="01:02:41:18" end="01:02:42:21" style="default" region="Region_Bottom">Dva mesiace.</p>
<p begin="01:02:43:06" end="01:02:47:22" style="default" region="Region_Bottom"><span tts:fontStyle="italic">Áno,
zabudla som číslo</span><br/><span tts:fontStyle="italic">tvojej skrinky. Povedal mi ho Chytrák.</span></p>
<p begin="01:02:48:17" end="01:02:50:24" style="default" region="Region_Bottom">Čo sa niekde stretnúť?</p>
<p begin="01:02:52:09" end="01:02:53:12" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Nemôžem.</span></p>
<p begin="01:02:54:07" end="01:02:58:03" style="default" region="Region_Bottom">-Prečo nie?<br/><span
tts:fontStyle="italic">-Ja som to pobabrala.</span></p>
<p begin="01:02:59:04" end="01:03:01:04" style="default" region="Region_Bottom"><span
tts:fontStyle="italic">Poriadne.</span></p>
<div>
<body>

```

FILE FORMAT SPECIFICATIONS (DATA) for Character Based Languages that may contain horizontal and vertical alignment concurrently (i.e., Japan)

- The Header information should be the similar (depending on the tool used to create) as the XML format example below:
- Example for 709 (HD/SDR):

```

<?xml version="1.0" encoding="utf-8"?>
<tt xmlns="http://www.w3.org/ns/ttml" xmlns:ttp="http://www.w3.org/ns/ttml#parameter"
xmlns:tts="http://www.w3.org/ns/ttml#styling" xmlns:ttml="http://www.w3.org/ns/ttml#metadata"
xmlns:ebutts="urn:ebu:tt:style" xmlns:ebuttm="urn:ebu:tt:metadata" xml:lang="ja"
ttp:contentProfiles="http://www.w3.org/ns/ttml/profile/imsc1.1/text" ttp:timeBase="media" ttp:frameRate="24"
ttp:frameRateMultiplier="1000 1001">
<head>
<metadata>
<ttml:title>GTMV0000102333_04000_SUB_JA-JA</ttml:title>
<ttml:desc>IMSC 1.1 Text Profile document created by Sferastudios</ttml:desc>
<ebuttm:conformsToStandard>urn:ebu:tt:distribution:2014-01</ebuttm:conformsToStandard>
<ebuttm:conformsToStandard>urn:ebu:tt:distribution:2014-01</ebuttm:conformsToStandard>
<ebuttm:conformsToStandard>http://www.w3.org/ns/ttml/profile/imsc1/text</ebuttm:conformsToStandard>
<ebuttm:conformsToStandard>http://www.w3.org/ns/ttml/profile/imsc1.1/text</ebuttm:conformsToStandard>
</metadata>

```

- The styling and layout information should appear as below:
- Example for 709 (HD/SDR)
- Font color is 92 grey 235,235,235 with legal range black outline 101010 at 3% thick. Font size differences highlighted as noted above.


```

<styling>
  <style xml:id="default" tts:color="#ebeb" tts:backgroundColor="transparent" tts:fontFamily="proportionalSansSerif"
  tts:fontSize="100%" tts:fontWeight="normal" tts:fontStyle="normal" tts:textOutline="#101010 3%"/>
</styling>
<layout>
  <region xml:id="Region_Top" tts:backgroundColor="transparent" tts:showBackground="whenActive" tts:origin="5%
  5%" tts:extent="90% 90%" tts:textAlign="center" tts:displayAlign="before" tts:luminanceGain="1.5"/>
  <region xml:id="Region_Bottom" tts:backgroundColor="transparent" tts:showBackground="whenActive"
  tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="center" tts:displayAlign="after" tts:luminanceGain="1.5"/>
  <region xml:id="Region_Vertical_Left" tts:backgroundColor="transparent" tts:showBackground="whenActive"
  tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="start" tts:displayAlign="after" tts:writingMode="tbrl"
  tts:luminanceGain="1.5"/>
  <region xml:id="Region_Vertical_Right" tts:backgroundColor="transparent" tts:showBackground="whenActive"
  tts:origin="5% 5%" tts:extent="90% 90%" tts:textAlign="start" tts:displayAlign="before" tts:writingMode="tbrl"
  tts:luminanceGain="1.5"/>
</layout>

```

- The position, luminance / color tag, and time base information etc. should be in the format below:
- Example for 709 (HD/SDR):
- **Note:** the default style and subsequent vertical alignment with shear adjustment example on a particular line if needed are highlighted for reference

```

</head>
<body>
  <div>
    <p begin="01:26:01:16" end="01:26:02:19" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">カ
    メラが先だ</p>
    <p begin="01:27:28:10" end="01:27:32:09" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">他
    人は私を<br/>トレジャーハンターと呼ぶ</p>
    <p begin="01:27:32:11" end="01:27:34:02" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">で
    も誤解よ</p>
    <p begin="01:27:34:05" end="01:27:36:14" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">金
    銭が目的じゃない</p>
    <p begin="01:27:37:15" end="01:27:40:11" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">私
    は錬金術の 歴史の研究者よ</p>
    <p begin="01:27:42:15" end="01:27:43:19" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">あ
    の石には&#11834;</p>
    <p begin="01:27:44:21" end="01:27:46:19" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">歴
    史的価値がある</p>
    <p begin="01:27:48:07" end="01:27:49:23" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">真
    実の探求が目的？</p>
    <p begin="01:27:50:07" end="01:27:51:07" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">そ
    うよ</p>
    <p begin="01:27:54:17" end="01:27:56:06" style="default" region="Region_Bottom" ebutts:multiRowAlign="start">真
    実が知りたい</p>
    <p begin="01:32:13:01" end="01:32:15:19" style="default" region="Region_Vertical_Right" tts:shear="16.67%">出るに
    は下りるしかない</p>
  </div>
</body>
</tt>

```


FILE FORMAT VALIDATION

The IMSC-1.1 file document shall be validated without error using the latest available version of the TTT--TTXV validator against the "imsc1.1" model.

- A web-based validator can be found at: <https://imscv.sandflow.com/>
 - Please select the IMSC 1.1 in the drop-down
- The tools use the imscJS library found at: <https://github.com/sandflow/imscJS>
- An experimental toolset to render the IMSC-1.1 file for visual verification can be found at: <http://sandflow.com/imsc1proc/index.html>

***NOTE: Reference the appropriate color coded header sections for additional component guidelines.**

FILENAMING CONVENTIONS

The table below outlines the required elements to be included in the filename. Please do not use any illegal characters in the file name i.e., : & , . () % # ~ œ / \ - { } [] < > ; @. Please note that the GTM Production Number field is equivalent to TVD Cross Reference Numbers.

DESCRIPTION	FILE NAME	EXAMPLES
Original Language Masters	[MAT ID]_[GTM Unified ID]_[GTM Production Number]_[Aspect Ratio + Transformation]_[Standard]_[Master Type]_[Language Code].extension	ABCD12345_GTMV0000123456_A1234_178FF_HD_TXM_EN-US.mov
International Masters	[SeriesTitle]_[Season#]_[Episode#]_EpisodeTitle_Framerate_[Aspect Ratio + Transformation]_[AudioConfiguration].extension	Leverage_S2_Ep204_TheBigRigJob_25i_16x9_178_Stereo_ME_51_1080.mxf
Localized Masters	[GTM Unified ID]_[GTM Production Number]_[Aspect Ratio + Transformation]_[Standard]_[Master Type]_[Language Code].extension	GTMV0000123456_A1234_185LB_SD_TLM_ES-LA.mov
Multi-Language Deliverables	[GTM Unified ID]_[GTM Production Number]_[Aspect Ratio + Transformation]_[Standard]_[Master Type]_[Language Code 1]_[Language Code 2].extension	GTMV0000123456_B5678_239SM_HDR_TXM_EN-US_PT-BR.mov
Subtitle Burned-In Masters	[GTM Unified ID]_[GTM Production Number]_[Aspect Ratio + Transformation]_[Standard]_[Master Type]_SUB_[Primary Language Code]_[Burned-in Subtitle Language Code].extension	GTMV0000123456_C9012_255ANA_SDR_TLM_SUB_EN-US_DE-DE.mov
Components	[GTM Unified ID]_[GTM Production Number]_[Component Type]_[Language Code].extension	GTMV0000123456_A1234_CC_EN-US.scc
Patch Fixes	[GTM Unified ID]_[GTM Production Number]_[Aspect Ratio + Transformation]_[Standard]_[Master Type]_[Language Code]_[Fix Number].extension	GTMV0000123456_A1234_185LB_SD_TLM_EN-US_FIX1.mov

Master Type Naming

MASTER TYPE	ABBREVIATION
Texted Master	TXM
Textless Master	TLM

Video Standard Naming

VIDEO STANDARD	ABBREVIATION
HD	HD
SD	SD
UHD HDR-10	UHD_HDR
UHD SDR	UHD_SDR
Dolby Vision	DV

Component Type Naming

COMPONENT TYPE	ABBREVIATION
Mono	MONO
Stereo	STE
Surround – Front Pair	LR
Surround – Center/LFE Pair	CLFE
Surround – Rear Pair	LRS
Viewer Access Satellite Television	VAST
Music	M
Effects	E
Dialogue	D
Full Mix	FM
Music & Effects	ME
Music & Dialogue & Effects	MDE
Closed Captioning	CC
Subtitle	SUB
Forced Narrative	FN
Voice Over	VO
Image	IMG
Dub Cards	DUBCARD

TechOps Inbound Specifications

Video Descriptive Service	VDS
Secondary Audio Programming	SAP
Audio Dub	DUB
Scripts	SCRIPT
Atmos Print Master	ATSPM
Atmos Music & Effects Master	ATSMEM

Language Codes Naming

LANGUAGE	ABBREVIATION
Albanian	SQ-AL
Arabic	AR-SA
Arabic (U.A.E.)	AR-AE
Armenian	HY-AM
Bosnian	BS-BA
Bulgarian	BG-BG
Catalan	CA-ES
Chinese (Simplified Mandarin, China)	ZH-CN
Chinese (Simplified Mandarin, Singapore)	ZH-SG
Chinese (Traditional Cantonese, Hong Kong)	ZH-HK
Chinese (Traditional Mandarin, Taiwan)	ZH-TW
Croatian	HR-HR
Czech	CS-CZ
Danish	DA-DK
Dutch (Std)	NL-NL
English (AUS)	EN-AU
English (CA)	EN-CA
English (South Africa)	EN-ZA
English (UK/GB)	EN-GB
English (US)	EN-US
Estonian	ET-EE

TechOps Inbound Specifications

Finnish	FI-FI
Flemish	FL-BE
French (Std)	FR-FR
French (Canadian)	FR-CA
German (Std)	DE-DE
Greek	EL-GR
Hebrew	HE-IL
Hindi	HI-IN
Hungarian	HU-HU
Icelandic	IS-IS
Indonesian	ID-ID
Irish	GA-IE
Italian (Std)	IT-IT
Japanese (Japan)	JA-JP
Kannada (India)	KN-IN
Korean (Korea)	KO-KR
Kurdish	KU-TR
Latvian	LV-LV
Lithuanian	LT-LT
Macedonian (FYROM)	MK-MK
Malay	MS-MY
Maltese	MT-MT
Maori	MI-NZ
Moldavian	RO-MD
Malaysian	BM-BM
Norwegian (Nynorsk)	NB-NO
Polish	PL-PL
Portuguese (Brazilian)	PT-BR
Portuguese (European)	PT-PT
Romanian	RO-RO
Russian	RU-RU
Serbian	SR-RS
Slovakian	SK-SK
Slovenian	SL-SI
Spanish (Ar)	ES-AR

TechOps Inbound Specifications

Spanish (Chile)	ES-CL
Spanish (Mexico, Latin America)	ES-MX
Spanish (Castilian)	ES-ES
Swedish	SV-SE
Tagalog	TL-PH
Tamil (India)	TA-IN
Telugu (India)	TE-IN
Thai	TH-TH
Turkish	TR-TR
Ukrainian	UK-UA
Vietnamese	VI-VN

NBCUniversal Quality Control Process

Quality Control

All deliverables must be checked for errors or defects to ensure compliance with the specifications listed within this document as well as NBCU content standards. If a QC report is available, please deliver as a PDF along with the other deliverables.

I. TECHNICAL QC

Upon delivery of audio, video, and component files to TechOps, a technical QC pass will occur. If there are any technical issues that require redelivery, Production will be contacted by the TechOps lead for the project to facilitate fixes and redelivery.

QC Rejections

The following is a list of common issues our QC teams see. If one of the issues below is encountered, it is considered a rejection and redelivery will be required.

VIDEO	AUDIO	VERSION & FORMATTING	COMPONENTS	OTHER
Video hits, digital artifacts, dead pixels, frame blending, visible macroblocking, banding, or inconsistent cadence	Audio out of range, ticks, dropouts, clipping, peaking, missing audio, sync errors, phase errors, unnatural or obvious ADR, etc.	Program not starting at hour one (01:00:00:00)	Components (closed captioning, subtitles, or forced narratives) are out of sync with video	Audio and video durations do not align (e.g. duplicate durations present on files)
Luminance, chrominance, or black levels out of range	Grouped audio delivered opposed to discrete audio featuring 1 track per channel	Incorrectly formatted (e.g. misplaced ad breaks) or containing extraneous materials (e.g. network bug, bumpers before/after ad breaks, website callouts, etc.)	Closed captioning or subtitle positioning covers actor faces, on-screen graphics, or on-screen text	Extraneous video or audio in the commercial blacks
Dolby Vision and HDR-10 files delivered without metadata		Textless material does not 100% match corresponding texted content		"Media Offline" banners or unrendered effects
Files delivered in multiple formats (HD, SD, UHD, Dolby, etc.) for one piece of content are not frame accurate to each other				

II. REDELIVERY

Less than 5 rejections may be sent as patch fixes directly to TechOps. Five or more rejections require re-delivery of the entire file. Please follow file naming guidelines for all re-deliveries and include information on timecode. Audio rejections should be addressed by the sound facility, followed by a full re-delivery of the entire package of audio deliverables.

III. THE HARDING TEST FOR PHOTOSENSITIVITY

Before providing content to NBCUniversal, content should undergo a Harding Test or Harding Analysis to check the program for instances of photosensitivity that could potentially cause seizures for viewers. Content may be subject to rejection if it fails to adhere to this regulatory standard. In the result of **failure**, the program must be **corrected** and **re-delivered** until the analysis results in a pass. Additionally, you must also **attach** a **photosensitivity warning** to the beginning of the content. Please contact your TechOps lead for access to the warning card.

Network, Cable & Studio Delivery Onboarding

Metadata

In addition to this spec document and the onboarding form, a metadata document will also be provided. Please complete the form including all required fields and send it back to TechOps in the .XLSX format. If you have any questions about filling out the metadata document, your Program Operations lead can help you.

The TechOps lead will provide a copy of the metadata template, which is comprised of the following sections:

I. TECHNICAL METADATA

- All fields marked as REQUIRED need to be filled out in order to procedurally generate the file names for asset delivery
- If delivering in HDR10, there are additional REQUIRED fields

II. FILE NAME COLUMNS

- These fields will auto-generate the required file names for the video, timed text, audio and image deliverables

File Transfer

The onboarding process concludes once test files have been received and validated by TechOps. Test file requirements will be discussed on the onboarding call prior to the file transfer. The **preferred** delivery method is via **Aspera P2P**, **Signiant P2P** or **Signiant Media Shuttle**.

If your chosen delivery method is **Aspera**, please provide the following:

- Aspera IP Address
- Username
- Password
- Source Directory Folder

If your chosen delivery method is **Signiant**, please provide the following:

- Signiant Agent
- Signiant Manager .XML for cross-trusting
- Source Directory Folder

If your chosen delivery method is **Signiant Media Shuttle**, please provide the following:

- Email Address (Maximum of 2 email accounts)

In order to set up Media Shuttle, please provide your TechOps lead with the email address to be used for account creation. You will receive an automated email to the address provided to activate your account and set up a password.

***NOTE:** Internet upload speed must be at least 100mbps to facilitate successful file delivery.

Technical Guidelines for Material

Longform Video Requirements

DESCRIPTION	REQUIREMENT
Video Wrapper	QuickTime .mov *NOTE: .mov files should contain no essence encryption or other encryption of any kind
Video Levels	White level luminance should not exceed 100 IRE/700mV when set to reference color bars, with peak offshoots up to 730mV. Black levels (setup) should not go below 0 IRE/0mV when set to reference color bars.
Frame Rate	Original, i.e., 23.98p, 29.97i, 29.97p, 25i *NOTE: 25p is accepted for Original
Aspect Ratio	Original, i.e., 1.78 Full Frame or Letterbox (2.35, 1.85, etc.)
Time Code	Time code track within wrapper. All 29.97 content must be Drop Frame. 23.98 and 25 content must be Non-Drop Frame
Color	No content can be scaled or tone mapped in any way (SDR --> HDR or HDR --> SDR). Content may be image scaled from HD --> UHD or from UHD --> HD with prior approval from TechOps.

I. HD SPECIFICATIONS

HD DELIVERABLE SPECS	
Video Codec	First Choice: ProRes 4:2:2 HQ Second Choice: XDCAM 50Mbps
Video Resolution	1920x1080
Color Primary	Rec 709
Color Tag	A QuickTime NCLC color tag is required for HD video so that it is properly identified in our servicing library. - Color Primaries = 1 - Transfer Function = 1 - Matrix Equation = 1

II. SD SPECIFICATIONS

SD DELIVERABLE SPECS	
Video Codec	First Choice: ProRes 4:2:2 HQ Second Choice: IMX 50
Video Resolution	720x486 (NTSC) 720x576 (PAL)
Color Primary	Rec 601
NTSC Color Tag	A QuickTime NCLC color tag is required for SD video so that it is properly identified in our servicing library. - Color Primaries = 6 - Transfer Function = 6 - Matrix Equation = 6
PAL Color Tag	A QuickTime NCLC color tag is required for SD video so that it is properly identified in our servicing library. - Color Primaries = 5 - Transfer Function = 6 - Matrix Equation = 5

III. UHD HDR-10 SPECIFICATIONS

UHD HDR-10 DELIVERABLE SPECS	
Video Codec	ProRes 4:4:4:4
Transfer Function	PQ
Video Levels	White level peak: 1000 nit Black level: .0001 nit
Video Resolution	3840x2160 UHD
Color Primary	Rec 2020 (constrained to P3 primaries)
Color Tag	A QuickTime NCLC color tag is required for HDR video so that it is properly identified in our servicing library. - Color Primaries = 9 - Transfer Function = 16 - Matrix Equation = 9
Metadata	Metadata will be required for all UHD HDR assets. Please populate in the metadata the appropriate fields: - MaxFALL - MaxCLL - WhiteLevelpeak(1000nit) - Blacklevel() (.0001 nit) - Transferfunction(PQ) - Primaries(P3)
SDR Delivery Requirement	For HDR-10 source native content delivered to TechOps an SDR file is required additionally to meet the delivery specifications of downstream platforms. TechOps will not convert HDR-10 files to SDR.

IV. UHD SDR SPECIFICATIONS

UHD SDR DELIVERABLE SPECS	
Video Codec	ProRes 4:2:2 HQ
Video Resolution	3840x2160 UHD
Color Primary	Rec 709
Color Tag	A QuickTime NCLC color tag is required for SDR video so that it is properly identified in our servicing library. - Color Primaries = 1 - Transfer Function = 1 - Matrix Equation = 1

V. DOLBY VISION SPECIFICATIONS

DOLBY VISION DELIVERABLE SPECS	
Video Codec	Preferred: ProRes 4:4:4:4 XQ (.mov) Acceptable: ProRes 4:4:4:4 12-bit (.mov)
Transfer Function	PQ
Video Levels	White level peak: 4000 nit (1000 nit if approved by TechOps) Black level: .0001nit
Video Resolution	3840x2160 UHD
Color Primary	Rec 2020 (limited to P3 primaries)
Color Tag	A QuickTime NCLC color tag is required for Dolby Vision video so that is properly identified in our servicing library. - Color Primaries = 9 - Transfer Function = 16 - Matrix Equation = 9
UHD- SDR Delivery Requirement	For Dolby Vision source native content delivered to TechOps a derived UHD/SDR file from the Dolby Vision 4.0 is required additionally to meet the delivery specifications of downstream platforms. If the Dolby version is 2.9 a separate UHD/SDR version is required, not created as a derived version utilizing the Dolby Vision 2.9 metadata TechOps will not derive Dolby Vision files to SDR.

Dolby Vision Guidelines

1. Please deliver a separate XML Dolby Vision Metadata file (version 4.0, Tech Ops is not currently accepting Dolby Vision 5.1 XML's) named the same as the content with the .xml file extension. The DVMD file should conform frame for frame to the delivered file, should not be shorter or

longer than the delivered file, and should contain all metadata changes at frame boundaries.

- When delivering a Dolby Vision File in Rec 2020 PQ, the original Dolby Vision Metadata has to be modified either with Metafier or manually to reflect the change of the file from P3 D65 to Rec 2020. The method to do this results in the changes below to the XML indicated in **BOLD and RED** only in the section "Main Layer" as highlighted. **NO** modifications should be done to any other "Primaries" section without speaking to Tech Ops. Please note if you have calculated max fall/cll for the HDR-10/1000 nit version of your content, please add those values to the section in **BOLD and BLUE**. If not, please leave as "0".

```
<Video>
  <Track name="MainLayer">
    <UniqueID>2</UniqueID>
    <Rate>
      <n>24000</n>
      <d>1001</d>
    </Rate>
    <ColorEncoding>
      <Primaries>
        <Red>0.708,0.292</Red>
        <Green>0.17,0.797</Green>
        <Blue>0.131,0.046</Blue>
      </Primaries>
      <WhitePoint>0.3127,0.329</WhitePoint>
      <MinimumBrightness>0.005</MinimumBrightness>
      <PeakBrightness>4000</PeakBrightness>
      <Encoding>pq</Encoding>
      <ColorSpace>rgb</ColorSpace>
      <SignalRange>computer</SignalRange>
      <BitDepth>16</BitDepth>
      <ChromaFormat>444</ChromaFormat>
    </ColorEncoding>
    <Level6>
      <MaxFALL>0</MaxFALL>
      <MaxCLL>0</MaxCLL>
    </Level6>
    <PluginNode>
      <DolbyEDR>
        <AlgorithmVersions>0,0</AlgorithmVersions>
        <Characteristics level="0"><MasteringDisplay id="7" name="4000-nit, P3, D65, ST.2084, Full">
          <ID>7</ID>
          <Name>4000-nit, P3, D65, ST.2084, Full</Name>
          <Primaries>
            <Red>0.68,0.32</Red>
            <Green>0.265,0.69</Green>
            <Blue>0.15,0.06</Blue>
          </Primaries>
          <WhitePoint>0.3127,0.329</WhitePoint>
          <MinimumBrightness>0.005</MinimumBrightness>
          <PeakBrightness>4000</PeakBrightness>
          <Encoding>pq</Encoding>
          <ColorSpace>rgb</ColorSpace>
          <SignalRange>computer</SignalRange>
          <DiagonalSize>42</DiagonalSize>
          <BitDepth>16</BitDepth>
          <ChromaFormat>444</ChromaFormat>
        </Characteristics>
      </DolbyEDR>
    </PluginNode>
  </Track>
</Video>
```

Audio Layouts

All audio must be discrete single-track audio (1 track per channel). The preferred layouts are listed below. If you plan to deliver Dolby Atmos audio, please work with your TechOps lead for further delivery instructions.

Music and Effects (M&E) are required for audio dubbing. Content cannot be dubbed by NBCUniversal without M&E being supplied via component or embedded audio.

NBCUniversal prefers 5.1 M&E to be delivered as component audio. See below for component delivery requirements.

CHANNEL	SURROUND + STEREO + M&E	STEREO + M&E	SURROUND + STEREO	STEREO	REALITY: TEXTLESS MASTER	SPORTS: STEREO	SCRIPTED: STEREO LAYOUT	SPORTS & SCRIPTED: 5.1 LAYOUT	TELEMUNDO: STEREO	TELEMUNDO: SURROUND
Ch1	5.1 Left (Front)	Stereo LT Primary Language	5.1 Left (Front)	Stereo LT Primary Language	Stereo LT Primary Language	Stereo LT Primary Language	Stereo LT Primary Language	5.1 Left (Front)	Stereo LT Primary Language	5.1 Left (Front)
Ch2	5.1 Right (Front)	Stereo RT Primary Language	5.1 Right (Front)	Stereo RT Primary Language	Stereo RT Primary Language	Stereo RT Primary Language	Stereo RT Primary Language	5.1 Right (Front)	Stereo RT Primary Language	5.1 Right (Front)
Ch3	5.1 Center	Stereo LT M&E	5.1 Center		Stereo LT M&E	Stereo LT Mix Minus	MOS	5.1 Center	MOS	5.1 Center
Ch4	5.1 LFE	Stereo RT M&E	5.1 LFE		Stereo RT M&E	Stereo LT Mix Minus	MOS	5.1 LFE	MOS	5.1 LFE
Ch5	5.1 Left (Rear)		5.1 Left (Rear)		Mono Voice Over	Mono Voice Over	MOS	5.1 Left (Rear)	MOS	5.1 Left (Rear)
Ch6	5.1 Right (Rear)		5.1 Right (Rear)		Mono Dialogue	Mono Natural Sound (SOT)	MOS	5.1 Right (Rear)	MOS	5.1 Right (Rear)
Ch7	Stereo LT Primary Language		Stereo LT Primary Language		Stereo LT Music	Stereo LT M&E	Stereo LT Primary Language	Stereo LT Primary Language	MOS	Stereo LT Primary Language
Ch8	Stereo RT Primary Language		Stereo RT Primary Language		Stereo RT Music	Stereo RT M&E	Stereo RT Primary Language	Stereo RT Primary Language	MOS	Stereo RT Primary Language
Ch9	Stereo LT M&E				Stereo LT Effects		Mono Dialogue	Mono Dialogue	Stereo LT M&E	Stereo LT M&E
Ch10	Stereo RT M&E				Stereo RT Effects		Mono Effects	Mono Effects	Stereo RT M&E	Stereo RT M&E
Ch11							Stereo LT Music	Stereo LT Music	Stereo LT Music	Stereo LT Music
Ch12							Stereo RT Music	Stereo RT Music	Stereo RT Music	Stereo RT Music
Ch13									Stereo LT Effects	Stereo LT Effects
Ch14									Stereo RT Effects	Stereo RT Effects
Ch15									Stereo LT Dialogue	Stereo LT Dialogue
Ch16									Stereo RT Dialogue	Stereo RT Dialogue

Technical Guidelines for Including Audio Description with Your Production:

Productions are to supply describer audio over a dual-channel mono version of the main soundtrack as follows:

- **Dual Mono Mix:** Create a dual-channel mono downmix bed of all elements of the stereo or 5.1 main soundtrack for AD mixing

- **Audio Ducking Process:** Ducking is typically applied by manual mixing, a console side-chain technique or an automatic microphone mixing device. Mix the audio describer on top of the dual-channel main mono downmix. Apply an appropriate amount of ducking to the dual-channel main mono downmix soundtrack that is triggered by the describer's modulation.
- **Audio Ducking Quality:** Ducking should be set in a fashion to emphasize the describer but not diminish the presence and impact of the background sounds. Ducking should not introduce any audible compression artifacts
- **Dialog Intelligibility:** The audio describer typically fits their commentary in between elements of the announcing or dialog of the main dual-channel mono downmix soundtrack. However, there may also be description that occurs at the same time as the dialog or commentary of the main dual-channel mono downmix soundtrack. Ducking should be applied to always create a level of good intelligibility of the describer dialog over the main mono downmix soundtrack
- **Dynamics:** The dual-channel mono composite audio description soundtrack (describer over main dual-channel mono downmix) should be managed for dynamics so good intelligibility and compliant loudness of the soundtrack are established and maintained
- **Loudness:** Foreground sound (whether the dialog of the describer or the dialog of the dual-channel main mono downmix) is to be provided at -24 LKFS average dialog loudness. This ensures the mono composite audio description soundtrack will match the loudness of interstitial content (ads, promos, etc.)
- **Output:** the dual-channel mono composite audio description soundtrack is to be delivered as a sidecar file recorded at -24 LKFS average dialog loudness, as noted above

Component Requirements

DESCRIPTION	REQUIREMENT
Closed Captioning Requirements	<i>Acceptable file types:</i> .scc Captions can include up to three lines of text per timecode and can only have 32 characters MAX per line. Note: If Italics are being used then the MAX characters per line is 31. Words either need to be moved to the next row (to have 3 rows on screen instead) or moved down to the next text box.
Audio	5.1 Surround, Stereo, or Dual Mono component audio is accepted Interleaved .wav file PCM 24-bit, 48 kHz
Subtitles	<i>Acceptable file types:</i> .html (IMSC1.1)
Other Scripts	<i>Acceptable file types:</i> .doc .docx *NOTE: As-Broadcast Scripts and/or Final Shooting Scripts are required for contextual tagging of incoming content.

Version & Formatting Requirements

DESCRIPTION	REQUIREMENT
Standard Text Version	Texted with Textless At Tail *NOTE: If the asset contains 20+ texted shots, it will be necessary to also provide a Fully Textless Master. (Shots mean actual shots with text in them. If there are multiple lines of text in one shot, that only counts as one texted shot).
Dolby Vision Text Version	Fully Texted AND Fully Textless *NOTE: Main and ends should remain texted.
Version Break Structure	Break Patterns (Episodic) Seamless (Theatrical)
Break Structure	1 second of black and silence between segments
File Start Time	00:58:30:00
Program Start Time	01:00:00:00

I. TIME FORMATTING & AD BREAKS

Total runtime should be calculated beginning at hour one (01:00:00:00) and ending precisely on the last frame of the master content, including ad breaks. Timecode should be continuous throughout the file, with no breaks, skips, or pauses in timecode data at any point. No time reduction or expansion processing shall be used by show producers that will affect audio, video and/or editorial presentation, including time compression, time tailoring and or vari-speeding.

The preferred start time and head-build construction is listed below and includes bars and tone and a descriptive slate. Some exceptions to this formatting may be permissible but must be approved by NBCUniversal. Start of content must be at 01:00:00:00 time code.

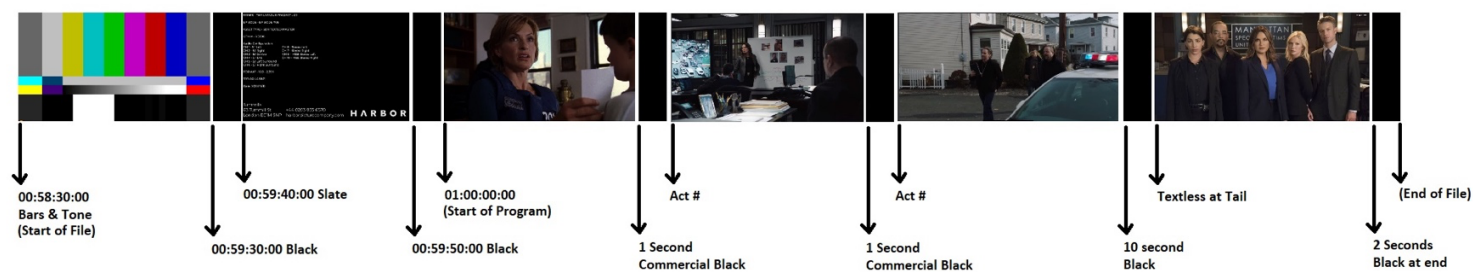
TIMECODE START:	DESCRIPTION	AUDIO	DURATION
00:58:30:00	Start of file, start of bars & tone	1kHz tone	60 seconds
00:59:30:00	Black	Digital silence	10 seconds
00:59:40:00	Slate	Digital silence	10 seconds
00:59:50:00	Black	Digital silence	10 seconds
01:00:00:00	Start of content	Program	Depends on program time

TechOps Inbound Specifications

-	Commercial Break (if applicable)	-	1 second
-	Act #	-	Depends on program time
-	Black before textless	-	10 seconds
-	Textless at tail slate (Optional)	-	3 seconds
-	Textless at tail material	-	Depends on program time
-	Black at end	-	2 seconds (minimum)

Timing Segment Format

Below is a visual representation of the segment format:



II. SLATE FORMAT

Please include the fields below on the slate. If you wish to include additional information, it may be added in supplement to these required items.

INFORMATION REQUIRED
TITLE OF PROGRAM
TITLE OF EPISODE (if applicable)
ASSET TYPE (Texted Master or Textless Master)
GTM PRODUCTION #/MAT ID
AUDIO TRACK ASSIGNMENT
FORMAT (Aspect Ratio + Transformation e.g., 16x9 Letterbox)
TOTAL RUNNING TIME